PIERRICK LORANG

Research Scientist • Robot Learning • Neuro-Symbolic AI • Autonomous Systems

Ph.D. Candidate, Tufts University • Mechanical Engineering & Human-Robot Interaction • Thesis: AI for Robot Learning • Expected May 2026

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Robot learning researcher specializing in getting robots to work reliably in the **real world**. 4+ years deploying RL + symbolic AI systems on physical hardware (robot arms, industrial forklifts, autonomous vehicles). **13 papers published or submitted in 4 years**, including 6 first-author at CoRL, ICRA, IROS & IJCAI. Core expertise: making robots generalize to novel tasks with minimal data, using foundation models to close the perception-to-action gap, and building systems that are both capable and interpretable. **Available June 2026.**

13	CoRL · ICRA×3	4+	≤5
Papers Published or Submitted (4 yrs)	IROS · IJCAI 6 First-Author Venues	Years on Real Hardware	Demos to Learn New Tasks

CORE COMPETENCIES

Robot Learning	Reinforcement Learning (model-based & model-free), Imitation Learning, Diffusion Policies, VLA fine-tuning
Planning & Reasoning	Neuro-symbolic integration, PDDL / Fast Downward, POMDP task planning, Epistemic planning, LLM-guided symbolic planning
Foundation Models	LLM/VLM (GPT, Claude, LLaMA, Gemini), Vision-Language-Action models (OpenVLA π_0, $\pi_{0.5}$), usage and tuning
Real-World Systems / Sim	Robot arms, industrial forklifts, CARLA autonomous driving, MuJoCo, Gazebo, Isaac Sim, ROS/ROS2, MoveIt
Perception	YOLO, Mask R-CNN, Vision Transformers (ViT/OWLv2/CLIP), OpenCV, semantic segmentation (SAM), pose estimation (FoundationPose)
Stack	Python, PyTorch, C++, CUDA, Docker, Singularity, Weights & Biases, Git, Linux, NumPy / Pandas

RESEARCH & INDUSTRY EXPERIENCE

Research Scientist — AI for Robotics | Austrian Institute of Technology & Tufts University HRI Lab *Oct 2021 – May 2026 • Vienna, AT & Medford, MA*

- Developed **few-shot imitation learning system** (CoRL 2025) enabling robots to acquire and generalize long-horizon manipulation tasks from **≤5 human demonstrations** — validated on Kinova arms; drastically reduces the data collection burden blocking real-world robot deployment.
- Built **curiosity-driven imagination framework** (ICRA 2025) where robots mentally simulate novel situations before acting, reducing real-world rollouts needed to discover viable policies on unseen tasks — directly addressing the sample efficiency bottleneck in robot RL.
- Deployed **LLM + VLM grounded symbolic planning pipeline** on industrial forklifts for automated pallet logistics — one of few academic groups to validate LLM-guided task planning on live industrial machinery.
- Engineered **hybrid hierarchical controller** for autonomous driving (CARLA) that maintained task performance under dynamic traffic and sensor degradation — symbolic planner handles high-level recovery while neural policy

executes low-level control.

- Designed **POMDP-based epistemic task planner** (IROS 2026, under review) enabling robots to detect and resolve knowledge gaps mid-task — critical for open-world deployment where environment state is never fully observable.
- Co-authored study (ICRA 2026) empirically showing **neuro-symbolic methods outperform VLAs on structured long-horizon tasks while consuming significantly less energy** — directly relevant to scalable, cost-efficient robot deployment.
- Published **6 first-author papers** (CoRL, ICRA, IROS, IJCAI-PRL) and **3 co-author papers** (AIJ, ICRA, IFAC) — **13 total papers published or submitted in 4 years**; 6 additional in submission or preparation for 2026 venues. Mentored M.Sc. and Ph.D. students; contributed to funded grant proposals.

AI / Computer Vision Engineer | Technology & Strategy — DataLab *Sep 2020 – Oct 2021 • Stuttgart, Germany*

- Designed and shipped **end-to-end production CV pipelines** (detection, segmentation, pose estimation) on PyTorch — from raw data to inference-ready models deployed on industrial datasets, with full preprocessing and scalable data pipelines.
- Tackled **low-label industrial data** via semi-supervised pseudo-labeling and aggressive augmentation, achieving meaningful performance gains where supervised baselines failed — practical experience in data-scarce real-world ML.

R&D Engineer — Autonomous Robotics | Capgemini / Altran *Mar 2020 – Oct 2020 • Toulouse, France*

- Built **autonomous drone navigation stack** (SLAM, 3D LiDAR + camera fusion, ROS/Gazebo) and multi-agent fleet architecture with human-machine handoff — early hands-on experience with full autonomy stacks before PhD.

Assistant Research Engineer — MEMS & Nanotechnology | Advanced Science Research Center, City College of New York (CUNY) *Sep 2018 – Feb 2019 • New York, NY*

- Designed and tested **high-sensitivity MEMS biosensors** for cancer cell detection in a state-of-the-art nanofabrication facility — demonstrates ability to operate in rigorous, precision-critical research environments well before PhD.

SELECTED PUBLICATIONS (FIRST-AUTHOR UNLESS NOTED)

- Duggan*, Lorang* et al., "Foundation Models vs. Compositional Reasoning for Long- Horizon Robotic Manipulation: An Energy and Performance Analysis," **ARR 2026.**
- Lorang et al., "Few-Shot Neuro-Symbolic Imitation Learning for Long-Horizon Planning and Acting," **CoRL 2025.**
- Lorang et al., "Curiosity-Driven Imagination: Discovering Plan Operators and Learning Associated Policies for Open-World Adaptation," **ICRA 2025.**
- Lorang et al., "Adapting to the "open world": The utility of hybrid hierarchical reinforcement learning and symbolic planning," **ICRA 2024.**
- Lorang et al., "A framework for neurosymbolic goal-conditioned continual learning in open world environments," **IROS 2024.**
- Lorang et al., "Speeding-up continual learning through information gains in novel experiences," **IJCAI 2022.**

EDUCATION

Joint Ph.D. — Mechanical Engineering & Human-Robot Interaction | Tufts University & Austrian Institute of Technology – GPA 4+/4 *2021 – 2026 (Expected) • Medford, US / Vienna, AT*

- Dissertation: **"Neuro-Symbolic Architectures for Open-World Robotic Adaptation"** — Advisor: Prof. Matthias Scheutz.
- Relevant coursework: Artificial Intelligence · Machine Learning · Reinforcement Learning · Probabilistic Robotics · Epistemic Planning · Graph Neural Networks · Generative AI.

M.Sc. Mechanical Engineering — Robotics & Mechatronics | Université de Technologie de Compiègne (UTC) — Sorbonne Universités – GPA 3.5/4 *2017 – 2020 • Compiègne, France*

- Erasmus exchange in Machine Learning & Mechatronics at Università di Bologna (2019–2020). Selected for Georgia Tech double-diploma program in ML.

LANGUAGES

French (Native) **English** (Fluent) **German** (Working Proficiency) **Italian** (Working Proficiency)